Jim van der Heijden

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Portfolio website | LinkedIn | Github

Projects

Two-Pass Meshlet HZB Occlusion Culling | DirectX 12, Mesh shaders, Slang, C++ 2026 | 8 weeks

- Implemented **GPU two-pass occlusion culling** per **meshlet**. Last frame's visible set seeds a conservative **Hi-Z depth pyramid**, and a late pass catches newly visible meshlets.
- Combined **frustum**, **normal-cone backface** and **Hi-Z occlusion** tests in a single **amplification-shader** pass, skipping hidden geometry before mesh shading.

Dynamic Diffuse Global Illumination using ray-traced irradiance fields | DirectX 12, C++ 2026 | 8 weeks

- Implemented **real-time dynamic diffuse global illumination** in a custom C++/DirectX 12 engine, using DXR to update a uniform 3D grid of irradiance probes every frame.
- Reduced light leaking with **Chebyshev-based probe visibility tests** on stored ray-hit distances, plus surface biasing and backface detection.

Godot Hybrid Hardware-Accelerated Ray Tracing | Vulkan, C++ 2026 | 10 weeks

- **Integrated a hybrid hardware-accelerated ray tracing pipeline into Godot's Vulkan renderer,** enhancing visual fidelity.
- **Adding a new rendering method** within a large, existing engine ensuring smooth integration.

Procedural grass rendering using indirect rendering | DirectX 12, C++ 2026 | 6 weeks

- **Researched and implemented a procedural grass rendering system** that enhances visual look while focusing on performance, handling many blades using indirect rendering techniques.
- **Built a procedural island generator** driven by height maps, with seamless integration into the grass rendering pipeline.
- **Ported an FFT-based ocean simulation** from the **PS5** and **added volumetric fog** to tie the scene together visually.

Nakon | Custom engine FPS game DirectX 12 and PlayStation 5, C++ 2025 | 16 weeks

- **Developed a custom DX12/PS5 renderer featuring deferred rendering, PBR, shadow mapping, HDR, post-processing and PS5 platform abstractions.**
- Built a **foliage system**, extended the **particle system**, and implemented post-process effects.
- Set up the **PlayStation 5 devkit build pipeline** and debugged **platform-specific issues**.

Skills

- Vulkan, PlayStation 5, DirectX 12
- C++, C#, HLSL
- Visual studio, CMake, Git, Perforce, Unreal engine, Unity, RenderDoc, PIX, Jira, Confluence
- Dutch (native/fluent), English (fluent)

Education

Breda University of Applied Sciences Netherlands, Breda
Creative Media and Game Technologies, Bachelors Programming 2023 - Present

SintLucas Netherlands, Eindhoven
Game developer, MBO-4 2019 - 2023

Work Experience

Meliora VR **Netherlands, Mierlo**

Developed VR interaction systems and gameplay features in Unity. **Jan 2023 - May 2023**